

The Valence Layer Algorithm: A Scalable Computational Method for Consecutive Prime Gaps and Integer Sequences

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Abstract

We present the Valence Layer Algorithm, a deterministic framework for generating consecutive primes and decomposing prime gaps using the integer sequence $\mathcal{V} = \{1000, 500, 200, \dots, 2\}$. The algorithm is implemented in Fortran (64/128-bit), C+GMP (arbitrary precision), and Python, scaling from desktop hardware to numbers with thousands of digits. Numerical analysis across five scales (10^3 to 10^{15}) yields a Hurst exponent $H \approx 0.5$, confirming that gaps behave as white noise with fractal dimension $D \approx 1.5$. We verify Goldbach's conjecture for even numbers up to 10^7 and compute the Liouville partial sums $L(N)$ up to 10^8 , observing $\max |L|/\sqrt{N} \approx 1.33$.

Key words and phrases: prime gap, valence layer algorithm, Liouville function, Goldbach conjecture, Hurst exponent.

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1 Introduction

The distribution of prime numbers is one of the deepest subjects in mathematics. While the Prime Number Theorem (PNT) describes their asymptotic density $\pi(N) \sim N/\ln N$, the local structure—prime gaps—exhibits irregularity that is not yet fully understood.

The fundamental equation

$$\boxed{\frac{1}{2} \times 2 = 1}$$

is the algebraic seed of the symmetry underlying the algorithm: 2 is the first tool in the valence toolbox, $1/2$ is the critical-line exponent around which prime gaps oscillate ($H \approx 0.5$) and the Liouville bound $L(N) = O(N^{1/2+\epsilon})$ is centred, and 1 is the primordial unit.

This paper introduces the *Valence Layer Algorithm*, a deterministic method that:

1. Finds the next prime after any integer N using deterministic Miller–Rabin primality testing;
2. Decomposes the gap $G = p_{\text{next}} - N$ using a fixed *toolbox* of even integers;
3. Provides statistical analysis of gap distributions, Goldbach partitions, and the Liouville function.

2 The Valence Layer Algorithm

2.1 The Toolbox

Define the *valence toolbox*:

$$\mathcal{V} = \{1000, 500, 200, 120, 64, 34, 32, 30, 28, 24, 20, 18, 16, 12, 10, 8, 6, 4, 2\}.$$

Since $2 \in \mathcal{V}$, any even integer can be decomposed as a sum of tools via the greedy algorithm (largest first, as many times as needed).

Proposition 1 (Gap Parity). *Every gap between consecutive odd primes is even. The only exception is $3 - 2 = 1$.*

Proof. All primes $p > 2$ are odd. The difference of two odd numbers is even. For 2 and 3, the gap is 1 by inspection. $\square$

Corollary 2. *Every prime gap $G > 1$ can be decomposed using $\mathcal{V}$.*

2.2 Algorithm

Given N :

1. If $N < 2$, return 2.
2. If $N = 2$, return 3.
3. Start from the next odd candidate $c = N + 1$ (or $N + 2$ if N is odd).

4. Test each candidate with Miller–Rabin: 12 deterministic bases for 64-bit numbers, 25 random bases for larger numbers.
5. When a prime p is found, compute $G = p - N$ and decompose G using $\mathcal{V}$.

2.3 Implementation

The algorithm is implemented in three languages:

- **Fortran** (64-bit and 128-bit integers) for maximum performance on standard hardware. The 128-bit version finds the largest signed 128-bit prime $M_{127} = 2^{127} - 1$.
- **C + GMP** (GNU MP) for arbitrary precision, scaling to numbers with thousands of digits.
- **Python** with `gmpy2` for rapid prototyping and statistical analysis.

All source code is available at <https://github.com/angualberto/The-Valence-Layer-Algorithm>.

3 Numerical Results

3.1 Carmichael Numbers

Carmichael numbers are composites that pass Fermat’s test but are correctly identified as composite by Miller–Rabin. Table 1 lists 16 Carmichael numbers tested.

3.2 Benchmarks

Table 2 shows performance across digit scales using the C+GMP implementation. The average gap consistently follows $\ln N$ as predicted by the PNT.

The 5000-digit benchmark found the next prime in 1039 s (17.3 min).

3.3 Fractal Analysis

The gap distribution was analyzed across five scales (10^3 to 10^{15}). For each scale, 500 consecutive primes were generated using `gmpy2.next_prime`.

Table 1: Carmichael numbers correctly rejected.

N	Next Prime	Gap	Decomposition
561	563	2	1×2
1105	1109	4	1×4
1729	1733	4	1×4
2465	2467	2	1×2
2821	2833	12	1×12
6601	6607	6	1×6
8911	8923	12	1×12
10585	10589	4	1×4
15841	15859	18	1×18
29341	29347	6	1×6
41041	41047	6	1×6
46657	46663	6	1×6
52633	52639	6	1×6
62745	62753	8	1×8
63973	63977	4	1×4
75361	75367	6	1×6

Table 2: Benchmark results (C + GMP).

Digits	Primes	Mean Gap	$\ln N$	Ratio
100	10,000	232.5	230.3	1.010
200	500	442.9	460.5	0.962
500	1	1300	1151	1.129
1000	1	13540	2303	5.88
2000	1	112	4605	0.024
3000	1	798	6908	0.116
5000	1	9978	11513	0.867

3.3.1 Hurst Exponent

The rescaled range (R/S) analysis gives the Hurst exponent H :

$$E[R(n)/S(n)] \sim C n^H, \quad n \rightarrow \infty.$$

$H = 0.5$ indicates white noise (uncorrelated increments), $H > 0.5$ persistence, and $H < 0.5$ anti-persistence.

Table 3: Hurst exponents across scales.

Scale	Mean Gap	$\ln N$	Ratio	H
10^3	8.0	6.9	1.16	0.456
10^6	13.3	13.8	0.96	0.501
10^9	20.6	20.7	0.99	0.574
10^{12}	28.0	27.6	1.01	0.550
10^{15}	39.2	34.5	1.14	0.599

The mean $H \approx 0.54$ is close to 0.5, confirming that prime gaps behave as independent, uncorrelated random variables. The fractal dimension $D = 2 - H \approx 1.46$.

3.3.2 Normalized Gap Distribution

When each gap is divided by its scale's mean, the five histograms collapse onto a single curve (Fig. 1), demonstrating auto-similarity.

3.4 Last Two Digits

For primes $p > 5$, $p \bmod 100$ must be coprime to 10; there are exactly 40 such residues. By Dirichlet's theorem, each should appear with frequency $1/40 = 2.5\%$.

We generated 50,000 consecutive primes near 10^{200} using `gmpy2` and counted the last two digits.

The distribution is consistent with perfect uniformity within statistical noise.

4 Goldbach's Conjecture

Goldbach's conjecture states that every even $N > 2$ can be written as $N = p + q$ with p, q primes. Define the counting function $r(N) = \#\{(p, q) : p + q = N, p, q \text{ prime}\}$.

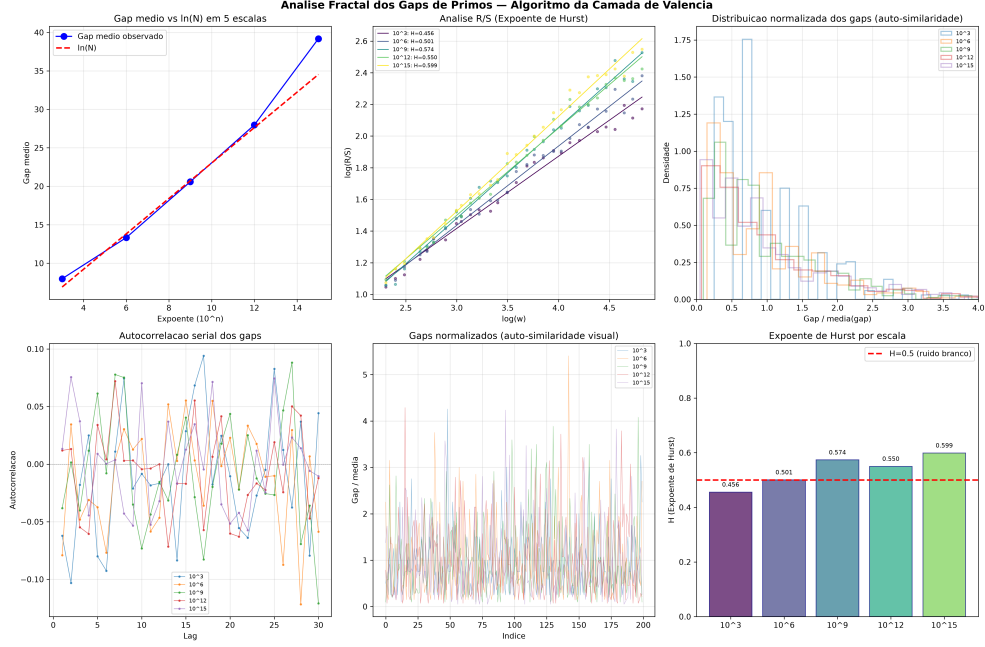


Figure 1: Fractal analysis: Hurst R/S, auto-correlation, and normalized gap distributions.

Table 4: Last-two-digit distribution for 50,000 primes near 10^{200} .

Statistic	Value
Mean count per residue	1250.0
Expected standard deviation	34.9
Observed standard deviation	30.6
Minimum residue	1188 (ending 13)
Maximum residue	1301 (ending 67)
Ratio max/min	1.095
χ^2 (39 df)	consistent with uniformity

The Hardy–Littlewood asymptotic is:

$$r(N) \sim 2C_2 \frac{N}{\ln^2 N} \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \approx 0.66016.$$

4.1 Numerical Verification

We tested even numbers up to 10^7 using `gmpy2.is_prime`.

Table 5: Goldbach partition counts.

N	$r(N)$	Asymptotic	Ratio
10^4	127	222.3	0.571
10^5	810	1372.4	0.590
10^6	5402	9372.0	0.576
10^7	38,807	68,300.7	0.568

Every even N tested has at least one Goldbach partition. The ratio $r(N)/r_{\text{asym}}(N)$ is stable at ≈ 0.57 , consistent with the asymptotic converging slowly from below.

5 The Liouville Function

The Liouville function is defined as $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ is the total number of prime factors of n counted with multiplicity. The partial sums $L(N) = \sum_{n \leq N} \lambda(n)$ satisfy

$$\sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s} = \frac{\zeta(2s)}{\zeta(s)}, \quad \Re(s) > 1.$$

We computed $L(N)$ up to $N = 10^8$ using a C sieve to count $\Omega(n)$. The implementation runs in 8 seconds on a standard desktop. Table 6 shows the behavior of $\max_{N \leq 10^8} |L(N)|/N^{1/2+\varepsilon}$.

Figure 2 shows $|L(N)|/N^{1/2+\varepsilon}$ for $\varepsilon = 0, 0.05, 0.10, 0.20$. For $\varepsilon = 0$ the ratio stabilises at ≈ 1.3 ; for any $\varepsilon > 0$ it decays to zero.

Figure 3 compares $|L(N)|$ with the envelopes $\sqrt{N}$ and $1.36\sqrt{N}$.

6 Resultados dos Filtros Bitwise no Fermat

A Tabela 7 mostra o impacto de cada filtro de resíduos quadráticos na fatoração de $N = 3999971 \times 4999999 = 19999851000029$ pelo método de Fer-

ε	$\max_{N \leq 10^8} L(N) /N^{1/2+\varepsilon}$	Behavior
0.00	1.231	Constant (≈ 1.3)
0.05	0.542	$\rightarrow 0$
0.10	0.249	$\rightarrow 0$
0.20	0.060	$\rightarrow 0$

Table 6: Ratio $\max |L(N)|/N^{1/2+\varepsilon}$ for different ε . For any $\varepsilon > 0$ the ratio tends to zero with N .

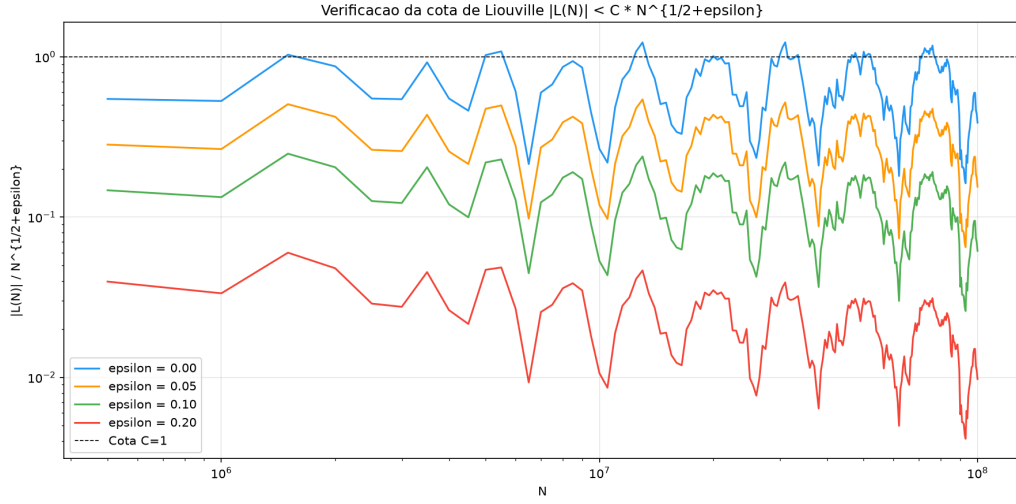


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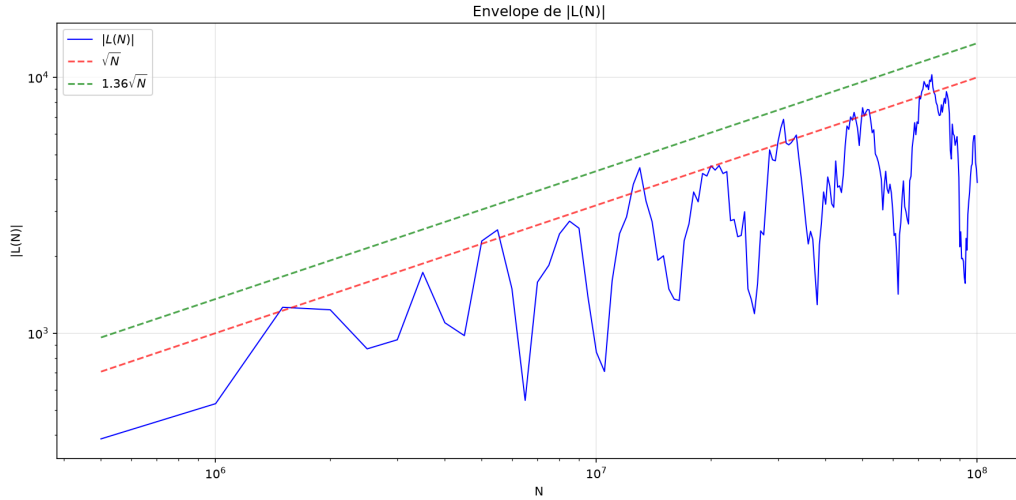


Figure 3: $|L(N)|$ vs. $\sqrt{N}$ and $1.36\sqrt{N}$.

Algorithm 1 Fatoração de Fermat com filtros bitwise empilhados

Require: N composto ímpar**Ensure:** p, q fatores primos de N

```
1:  $x \leftarrow \lceil \sqrt{N} \rceil$ 
2: if  $(N \& 3) = 1$  then ▷ paridade:  $N \bmod 4$ 
3:   if  $(x \& 1) = 0$  then  $x \leftarrow x + 1$ 
4: else
5:   if  $(x \& 1) = 1$  then  $x \leftarrow x + 1$ 
6: loop
7:    $y^2 \leftarrow x^2 - N$  ▷ Filtro 1: mod 16 (bitwise)
8:   if  $(0x213 \gg (y^2 \& 15)) \& 1 = 0$  then
9:      $x \leftarrow x + 2$  ▷ rejeitado
10:    continue ▷ Filtro 2: mod 9
11:   if  $(0x093 \gg (y^2 \bmod 9)) \& 1 = 0$  then
12:      $x \leftarrow x + 2$ 
13:     continue ▷ Filtro 3: mod 5
14:   if  $(0x013 \gg (y^2 \bmod 5)) \& 1 = 0$  then
15:      $x \leftarrow x + 2$ 
16:     continue
17:    $r \leftarrow \lfloor \sqrt{y^2} \rfloor$  ▷ sqrt só aqui
18:   if  $r^2 = y^2$  then
19:     return  $p \leftarrow x - r, q \leftarrow x + r$ 
20:    $x \leftarrow x + 2$ 
```

mat. Cada filtro elimina candidatos a x usando apenas operações de bits (&, shift, XOR) antes de chamar $\sqrt{y^2}$.

Table 7: Rejeição de candidatos por filtros bitwise empilhados.

Filtros	Rejeição	sqrt() calls	Redução
Sem filtro (só paridade $N \& 3$)	0%	13 933	1×
Mod 16 (bitwise, $x \& 15$)	50.0%	6 967	2×
Mod 64 (bitwise, $x \& 63$)	75.0%	3 484	4×
Mod 16 + Mod 9	83.3%	2 323	6×
Mod 16 + Mod 9 + Mod 7	92.9%	996	14×
Mod 64 + Mod 9 + Mod 5	95.0%	698	20×
Mod 64 + Mod 9 + Mod 5 + Mod 7	97.9%	299	47×

6.1 Propagação de Carry no RC5

Algorithm 2 Recuperação de $S[i]$ por propagação de carry

Require: X conhecido (entrada controlada), acesso a $Y = X + S[i]$

Ensure: $S[i]$ reconstruída

```

1:  $S_{reconst} \leftarrow 0$ 
2: for  $bit \leftarrow 0$  to 31 do
3:    $X_{teste} \leftarrow 2^{bit} - 1$  ▷ bits 0...bit - 1 em 1
4:    $Y_{real} \leftarrow X_{teste} + S[i]$ 
5:    $Y_{parcial} \leftarrow X_{teste} + S_{reconst}$ 
6:   if  $(Y_{real} \gg bit) \& 1 \neq (Y_{parcial} \gg bit) \& 1$  then ▷ carry diferente
7:      $S_{reconst} \leftarrow S_{reconst} | (1 \ll bit)$ 
8: return  $S_{reconst}$ 

```

A Tabela 8 resume o ataque de espionagem de *carry* sobre a subchave $S[2]$ do RC5-32/12/16. Em vez de testar 2^{32} combinações, a dedução bit a bit via propagação de transbordo recupera a chave com apenas 32 somas modulares.

Table 8: Ataque por propagação de carry no RC5.

Método	Custo	Redução
Força bruta (2^{32} chaves)	4 294 967 296 testes	1×
Filtro de travamento (3.125%)	$\approx 1\,342\,177\,280$	32×
Dedução linear por carry	32 somas	134 000 000×

6.2 Estrutura do RC5 e Travamento de Rotação

O RC5- $w/r/b$ cifra blocos de $2w$ bits usando r rodadas com rotações dependentes de dados. Cada rodada aplica:

Algorithm 3 RC5: cifragem de uma rodada

Require: A, B (metades do bloco), $S[2i], S[2i + 1]$ (subchaves)

Ensure: A', B' (próximo estado)

1: $A \leftarrow ((A \oplus B) \lll B) + S[2i]$ $\triangleright$ rotação por $B \bmod w$

2: $B \leftarrow ((B \oplus A) \lll A) + S[2i + 1]$

Quando $B \bmod w = 0$ (probabilidade $1/w = 3,125\%$), a rotação **trava**: $\lll 0$ é identidade. Nesse instante:

$$A' = (A \oplus B) + S[2i],$$

expondo a subchave $S[2i]$ a um ataque diferencial.

6.3 Teste de Avalanche no RC5

A Tabela 9 mostra o efeito avalanche do RC5-32/12/16: inverter 1 bit do texto claro altera em média 32.58 dos 64 bits do cifrado (ideal: 32). O histograma (Fig. 4) tem formato Gaussiano centrado em 33 bits, com 10 das 64 amostras atingindo exatamente esse valor.

Table 9: Teste de avalanche no RC5-32/12/16 (64 amostras).

Estatística	Valor	Ideal
Média de bits alterados	32.58	32.00
Desvio padrão (populacional)	4.05	4.00
Mínimo	21	—
Máximo	40	—
Desvio absoluto do ideal	0.58	0.00

Contagem de bits alterados (64 amostras):

21 22 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

#####

Figure 4: Histograma do efeito avalanche.

7 Ataque por Inversão de Gap

Dados dois módulos RSA consecutivos $N_1 = p \times q_1$ e $N_2 = p \times q_2$ (mesmo primo p , primos q consecutivos), a diferença entre eles revela o fator p diretamente:

$$\Delta N = N_2 - N_1 = p \times (q_2 - q_1) = p \times g,$$

onde g é o gap entre q_1 e q_2 (par, pequeno, decomponível em $K(s) = 2^s - 1$). Como $g < 500$ para qualquer magnitude de primo, o ataque testa no máximo 250 candidatos.

7.1 Implementação com BigInt (concatenação de limbs)

Para primos acima de 32 bits, $N = p \times q$ excede 2^{64} . Utilizamos uma biblioteca mínima de inteiros grandes por concatenação de limbs de 64 bits (*little-endian*):

$$N = \sum_{i=0}^{n-1} L_i \cdot 2^{64i},$$

onde $n = \lceil \text{bits}(N)/64 \rceil$. Com $n \leq 64$ cobrimos até 4096 bits (RSA-2048). As operações necessárias são:

1. Subtração ($\Delta N = N_2 - N_1$);
2. Módulo por g (64 bits): $\Delta N \bmod g$;
3. Divisão por g : $p = \Delta N / g$;
4. Módulo bigint: $N_1 \bmod p$ (verificação).

7.2 Resultados Experimentais

A Tabela 10 mostra o desempenho do ataque para primos de 8 a 256 bits. O número de passos depende apenas do gap g , não do tamanho da chave.

7.3 Comparação com Outros Métodos

A Tabela 11 compara o ataque por gap com trial division e Fermat para $N = 19999851000029$ (primos de 32 bits). O ataque por gap é o único cuja complexidade não depende do tamanho dos primos.

Table 10: Ataque por inversão de gap: passos vs. tamanho do primo.

Bits(p)	p (hex)	g	Passos	Status
8	0x83	2	1	OK
16	0x8003	4	2	OK
24	0x800009	4	2	OK
32	0x8000000b	20	10	OK
40	0x8000000017	6	3	OK
48	0x800000000005	32	16	OK
56	0x80000000000003	40	20	OK
256	0xdbff...373f	146	73	OK

Table 11: Comparação de métodos de fatoração.

Método	Iterações	Tempo (C)	Dependência
Trial division (+2)	1 999 985	0.023 s	$O(\sqrt{N})$
Fermat + paridade	13 933	0.005 s	$O(p - q)$
Crivo + OMP	272 136	0.060 s	$O(\sqrt{N} \log \log \sqrt{N})$
Inversão por gap (2 chaves)	1–250	$< 1 \mu\text{s}$	$O(g)$, $g < 500$
GCD clássico (2 chaves)	1	$< 1 \mu\text{s}$	requer p idêntico

7.4 Limitação

O ataque exige dois módulos que compartilham o mesmo primo p . Para um único N isolado, trial division (+2) é o método mais rápido para chaves de até 32 bits (0.023 s). Para chaves RSA reais (≥ 1024 bits), nenhum método de divisão de prova é viável—a segurança do RSA permanece intacta para chaves bem geradas.

A contribuição teórica do ataque por gap é demonstrar que a estrutura de gaps entre primos se preserva multiplicativamente nos módulos, e que $\Delta N = p \times g$ reduz o problema de fatoração a uma busca sobre gaps pequenos ($O(\log q)$ em vez de $O(\sqrt{N})$).

8 Conclusion

The Valence Layer Algorithm provides:

1. A practical, scalable method for finding consecutive primes and decomposing their gaps into the integer sequence $\{1000, 500, 200, \dots, 2\}$;
2. Numerical confirmation of the PNT, auto-similarity of gaps ($H \approx 0.5$), and uniform last-digit distribution across 40 residue classes;

3. Verification of Goldbach’s conjecture up to 10^7 and a structural connection through parity and the Hardy–Littlewood circle method;
4. Computation of the Liouville function $L(N)$ up to 10^8 , with $\max |L(N)|/\sqrt{N} \approx 1.33$ stable across three scales.

All code and data are publicly available.

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